Conformational analysis and Molecular orbital Using Avogadro and ORCA

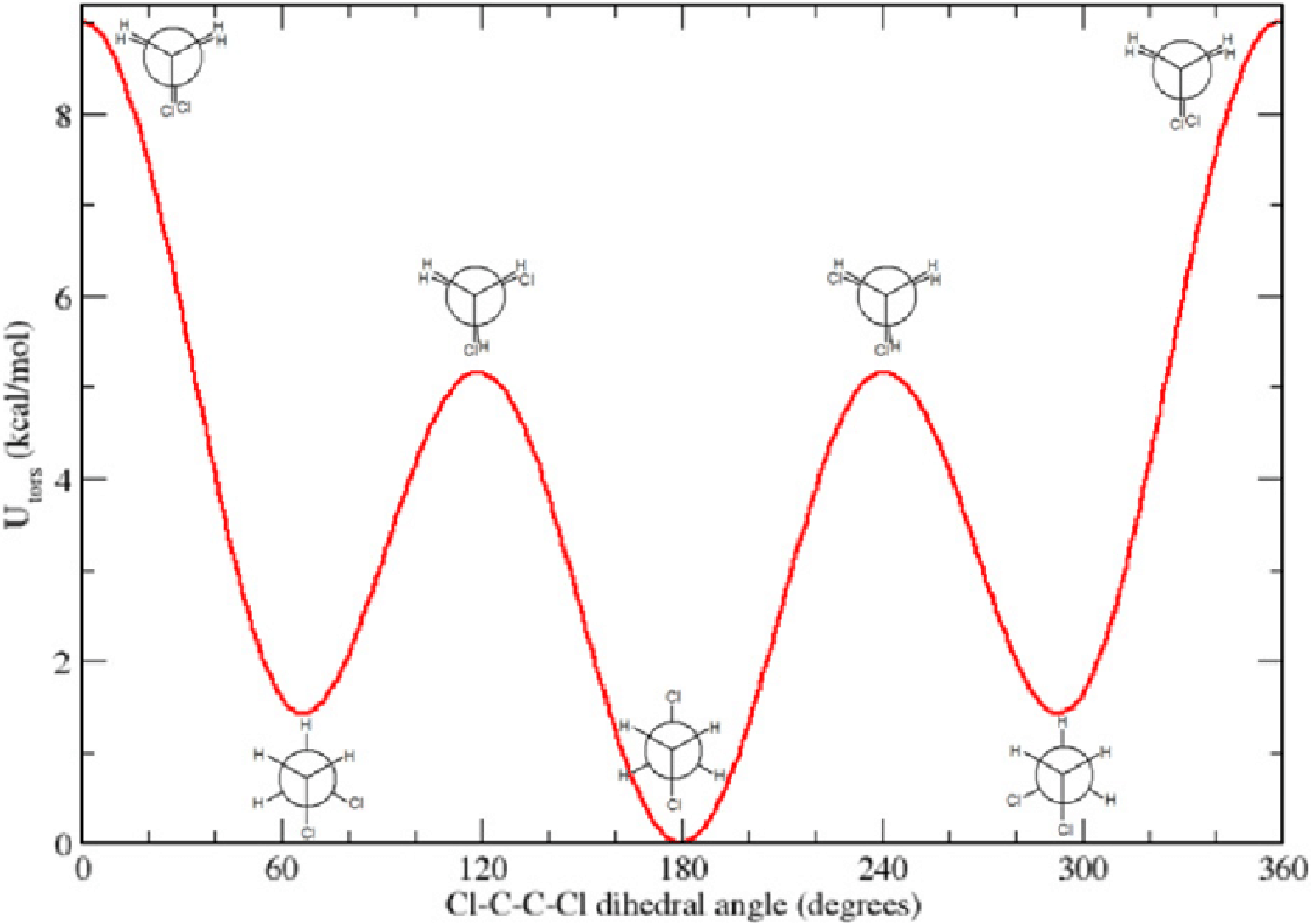
Experiment # 2

Format of Reporting Data

Structure		Energy	Bond length	Bond Angle	Torsion Angle
Molecule Before optimization	Molecule After optimization	Before optimization	Before optimization	Before optimization	Before optimization
		After optimization	After optimization	After optimization	After optimization

1,2-dichloroethane

- Change the dihedral angle and measure the energy
- Plot energy versus dihedral angle



U_tors (kcal/mol)
Cl-C-C-Cl dihedral angle (degrees)
0
2
4
6
8
60
120
180
240
300
360
H
Cl

Dimethyl cyclohexane isomers

- Cis-(axial, axial)-1,3-dimethyl cyclohexane (start from cyclohexane chair form)
- Trans- (axial, equatorial)- 1,3-dimethyl cyclohexane
- Cis-(equatorial, equatorial)-1,3-dimethyl cyclohexane

Change in energy between equatorial and axial conformation

- Methyl Cyclohexane
- T-Butyl Cyclohexane
- Trifluoro methyl cyclohexane

2,2-dimethyl 3-ethyl 5,5-dimethyl hexane

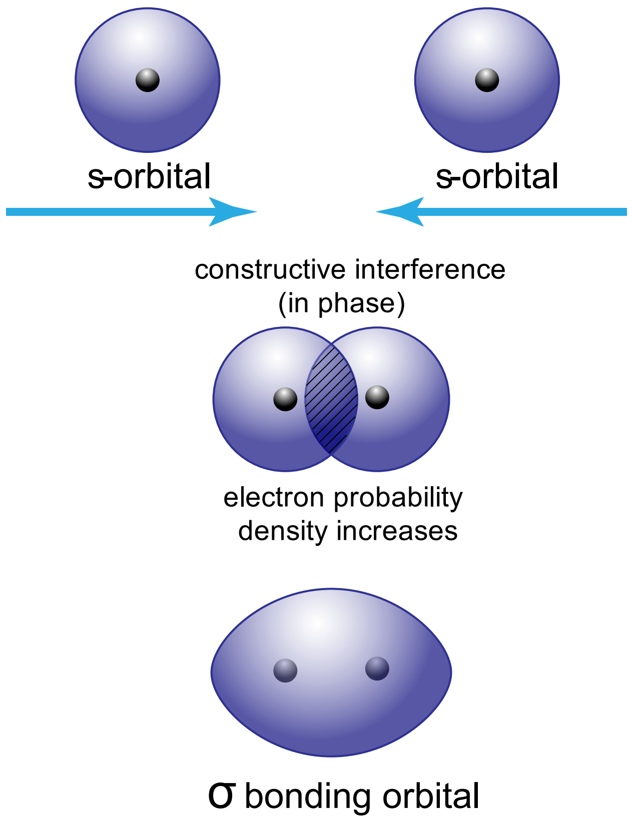
- Rotate about C3-C4 bond with different angle.
- Minimize and note down the energy
- View along C3-C4 bond.

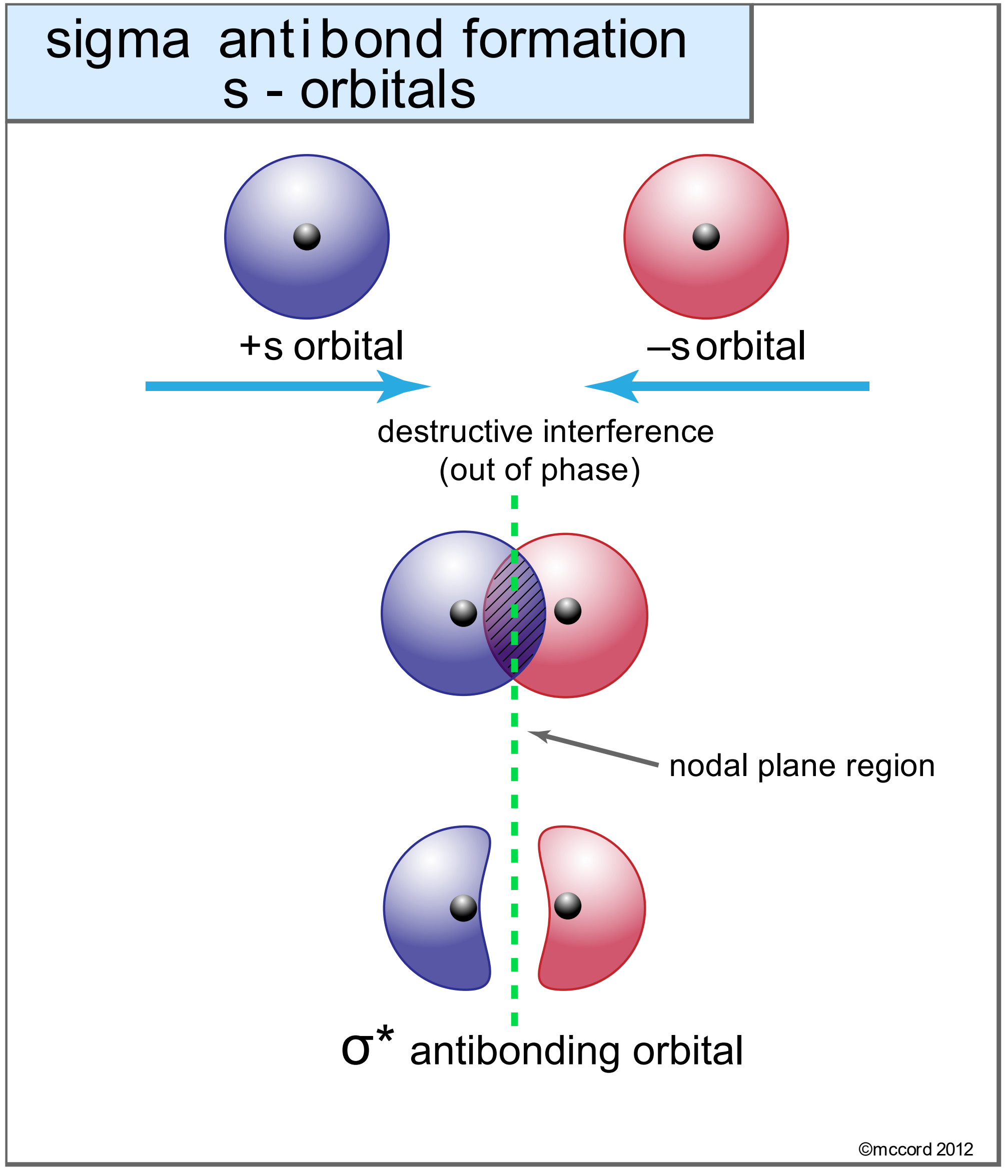
Optimization using ORCA software

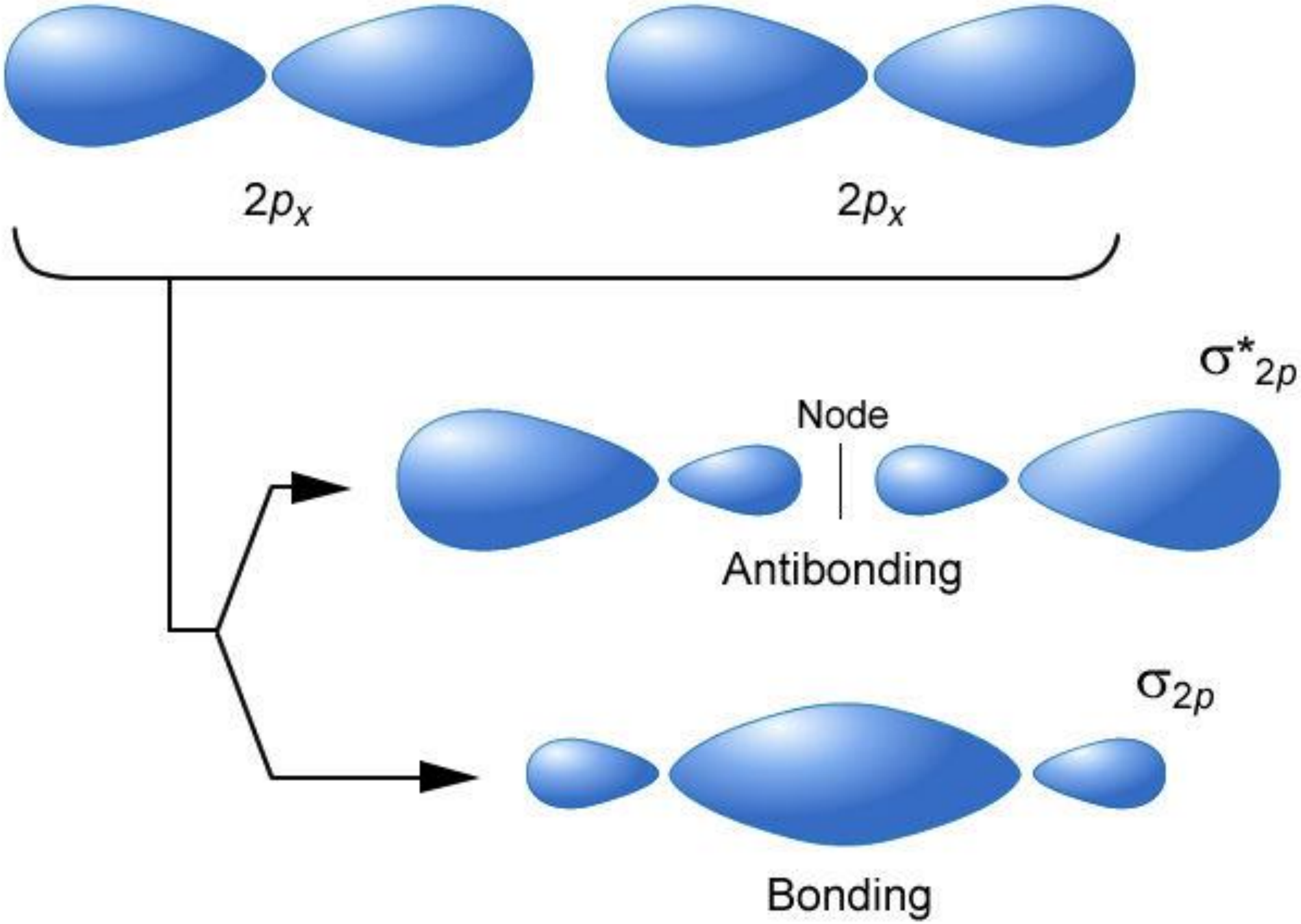
	VSEPR-predicted Structures	***Part A1 – Avogadro Optimization Results***			***Part C – ORCA optimization Results***		
		Energy	**Bond Lengths**	**Bond Angles**	**Energy & Time**	**Bond Lengths**	**Bond Angles**
PART A1	CH_4 ORCA Filenames:						
	C_2H_6 ORCA Filenames:						
	C2H4 ORCA Filenames:						

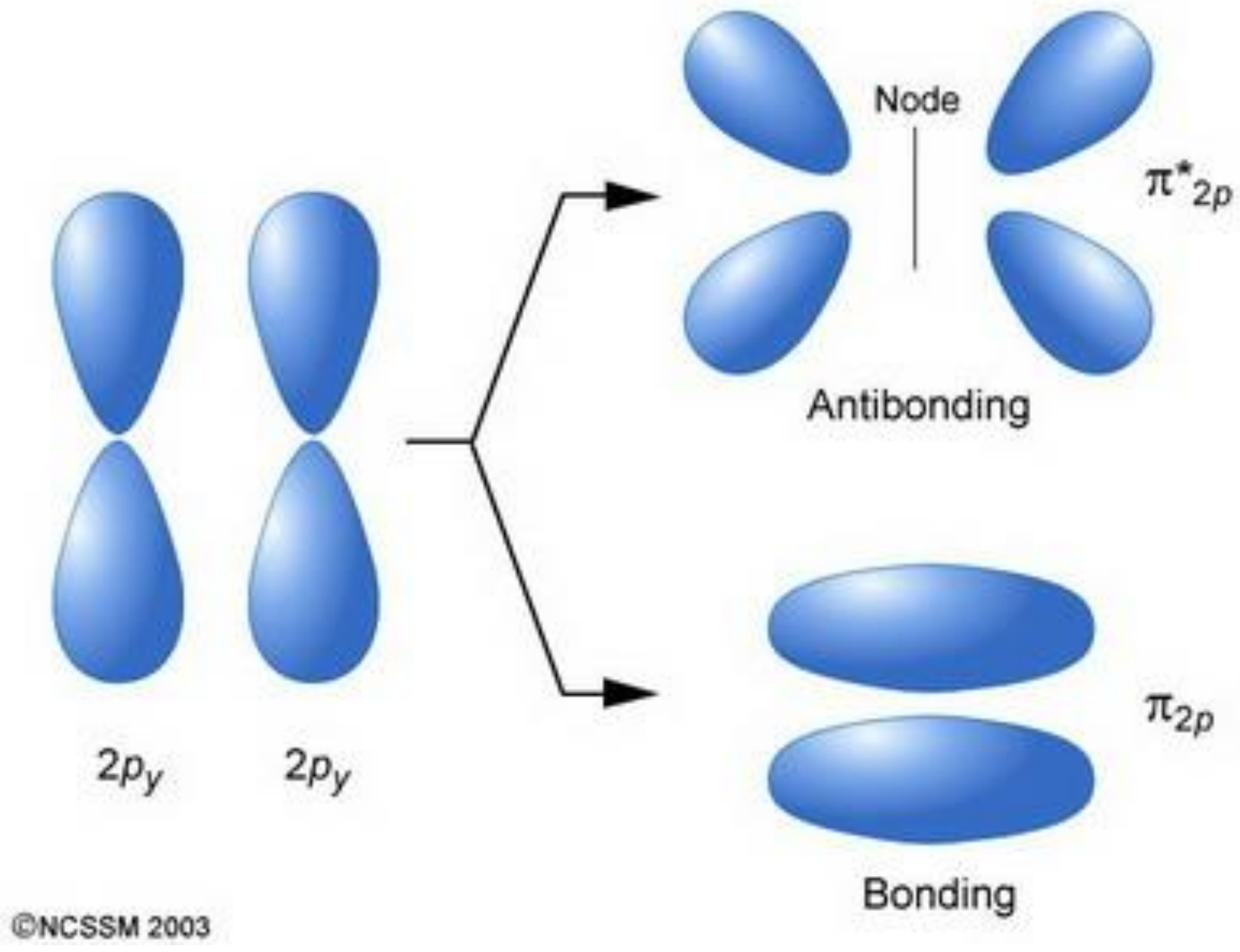
Formation of Molecular Orbitals

Sigma Bond Formation
s - orbitals




sigma antibond formation
s - orbitals
+s orbital
–s orbital
destructive interference
(out of phase)
nodal plane region
σ* antibonding orbital
©mccord 2012


$2p_x$
$2p_x$
σ^*_{2p}
Node
Antibonding
σ_{2p}
Bonding



Node
π^*_{2p}
Antibonding
π_{2p}
Bonding
$2p_y$
$2p_y$
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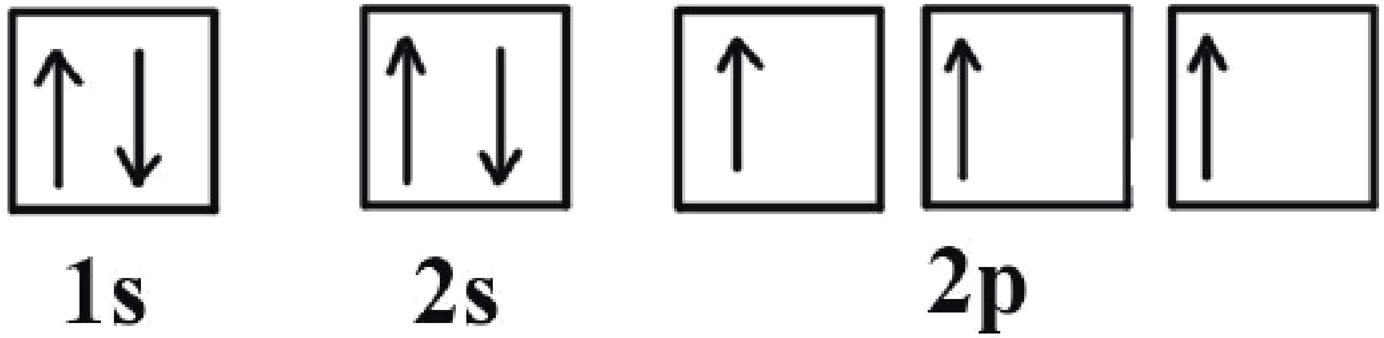
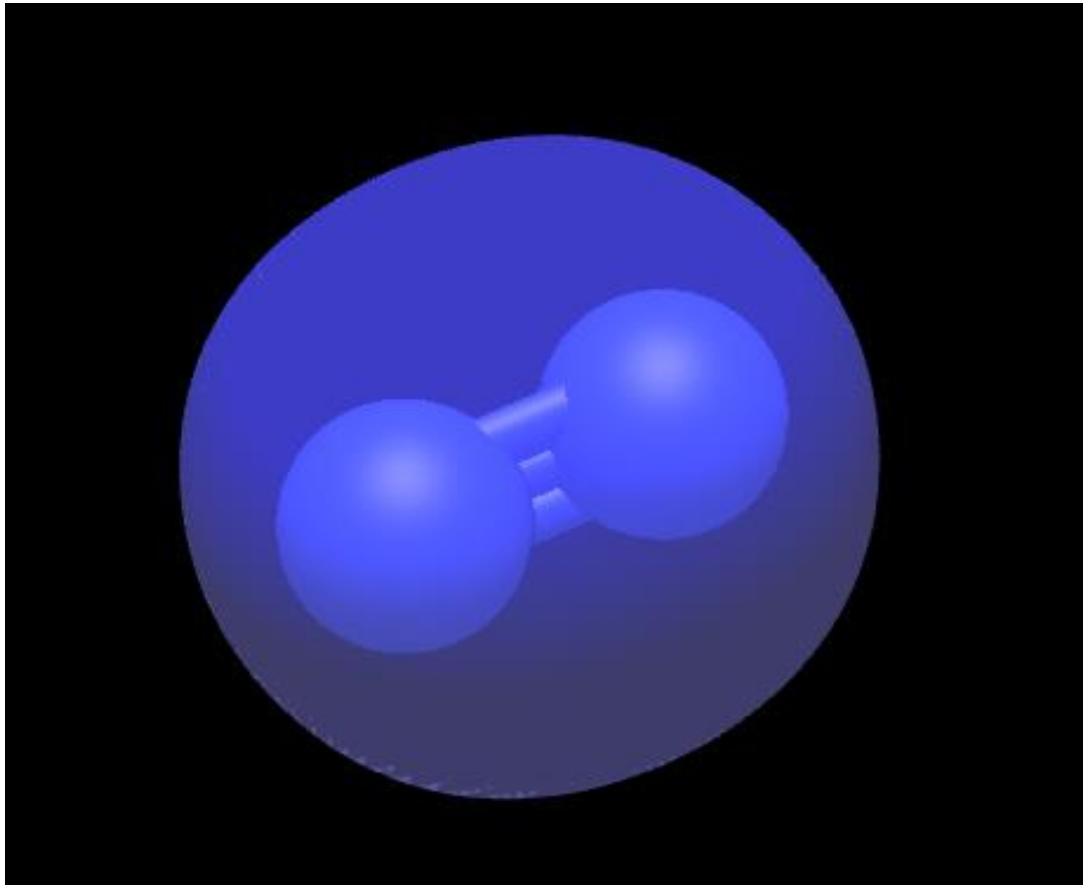
Molecular Orbitals of N_2 using ORCA and their energy

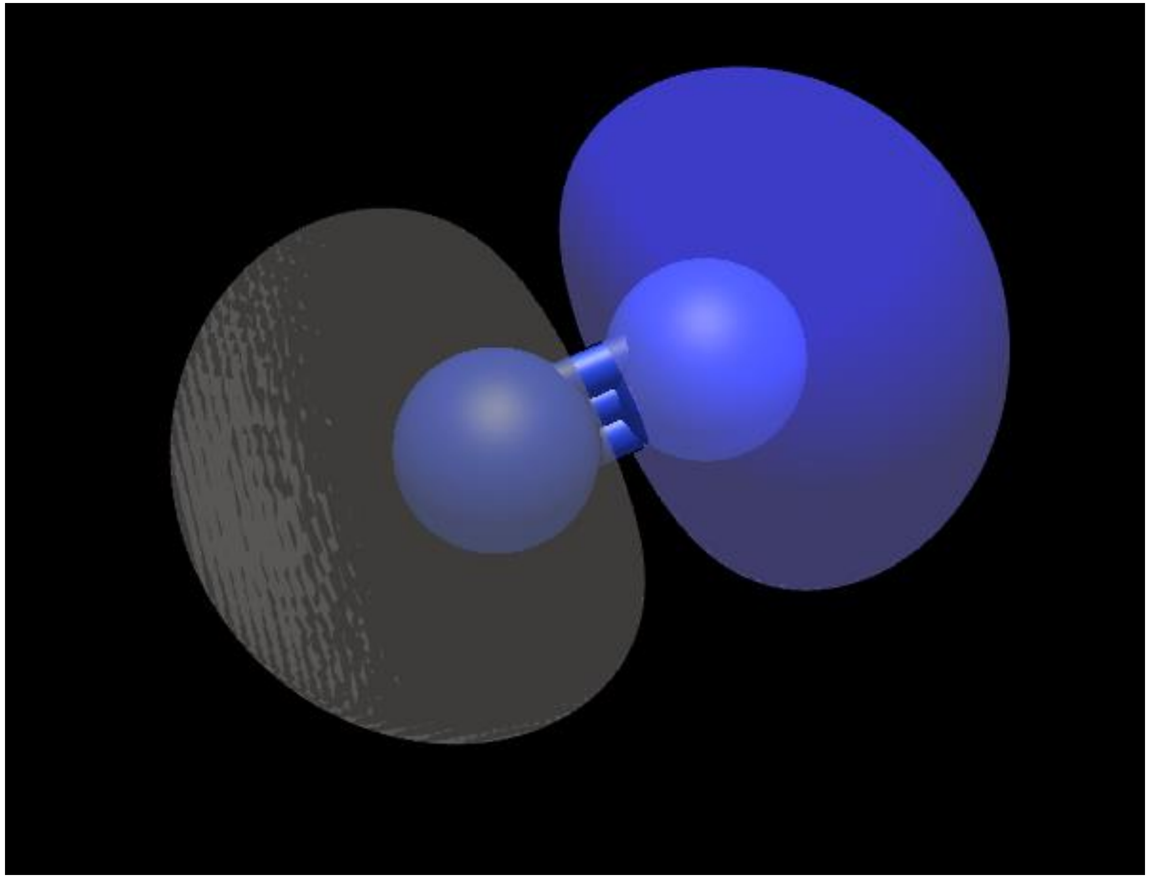
Nitrogen



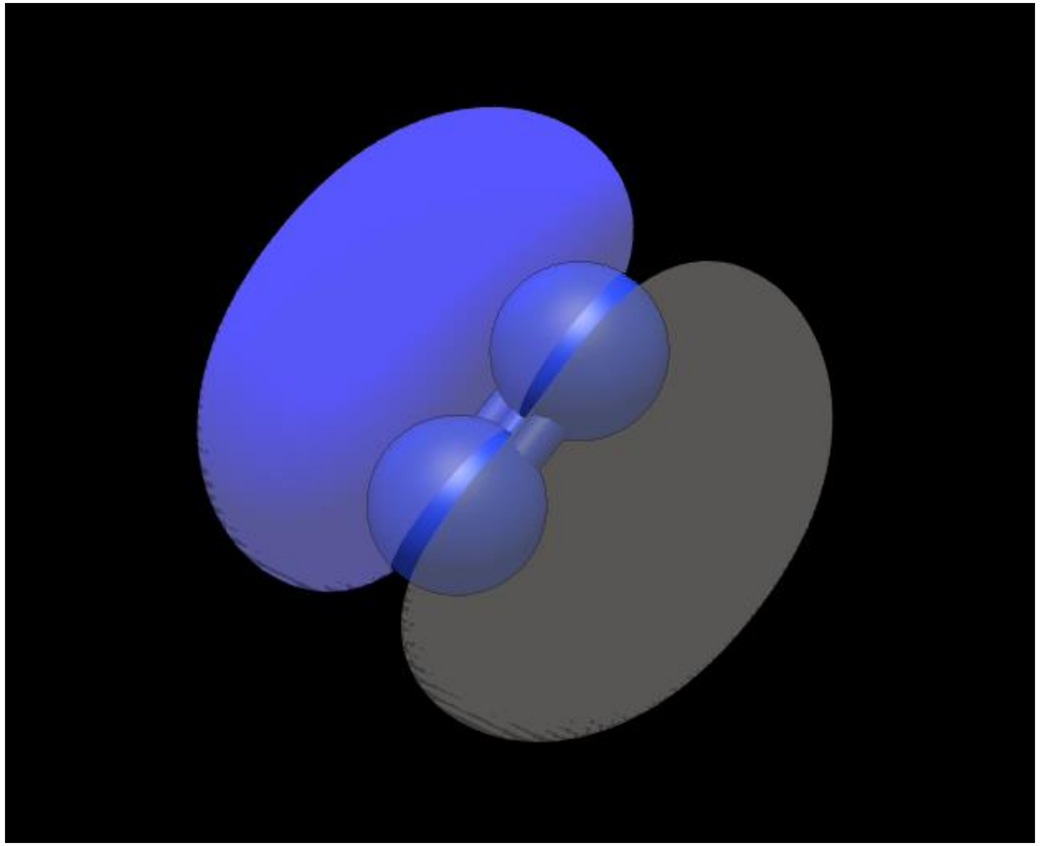
Sigma

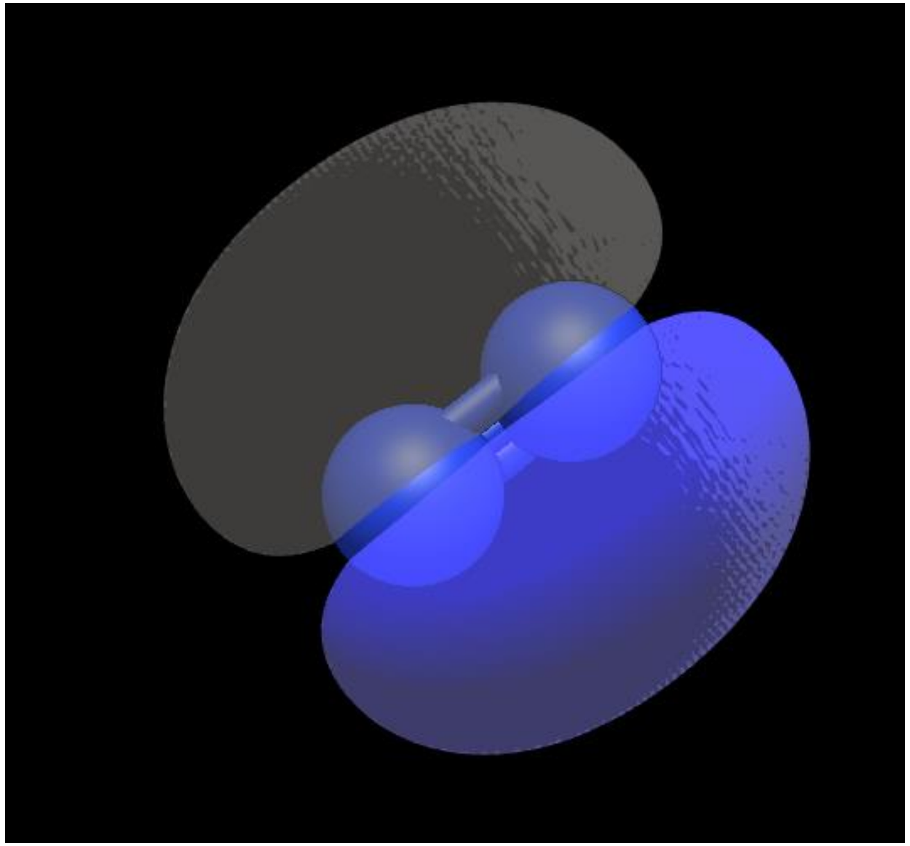
Sigma *



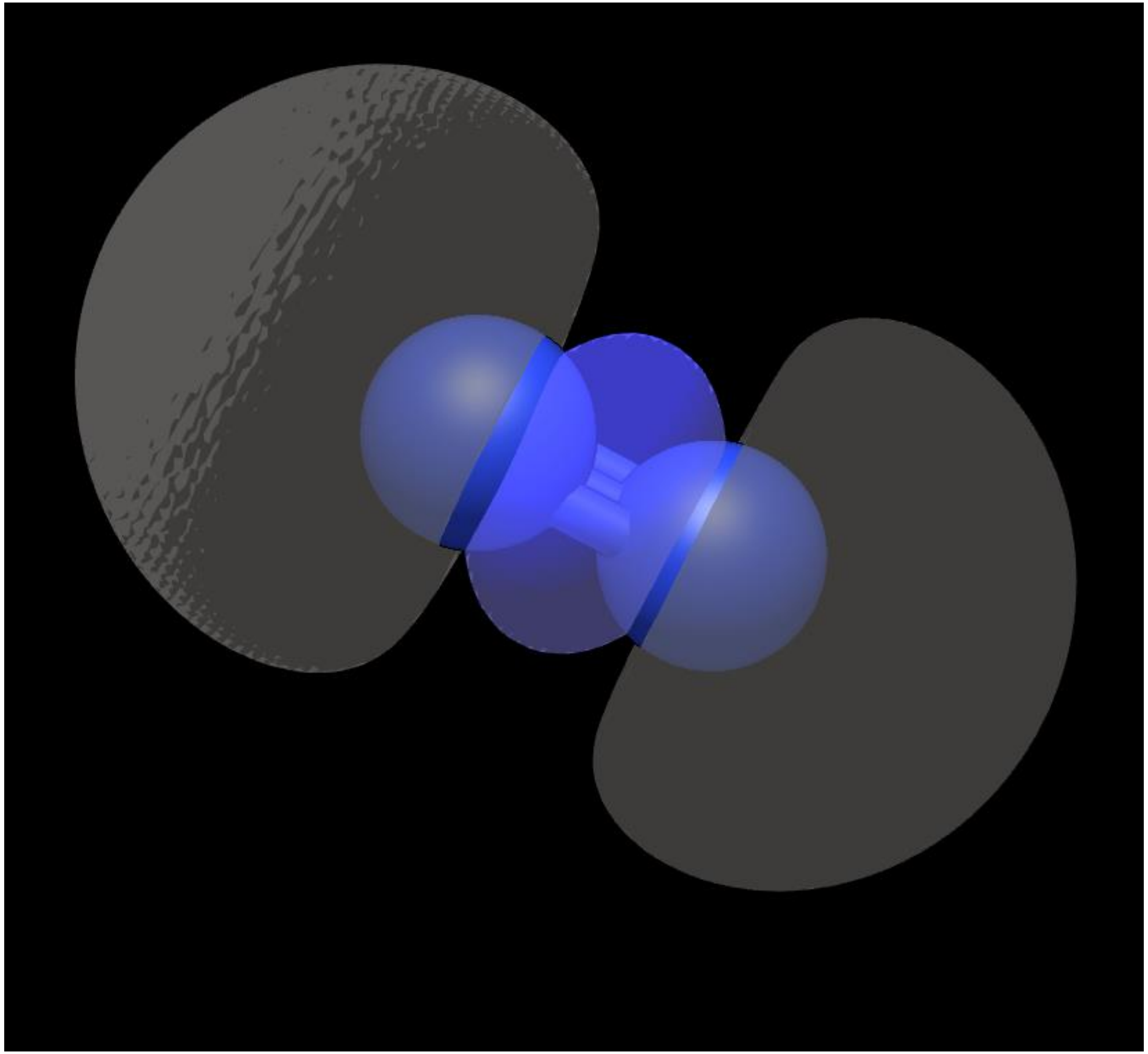


Pi 2p

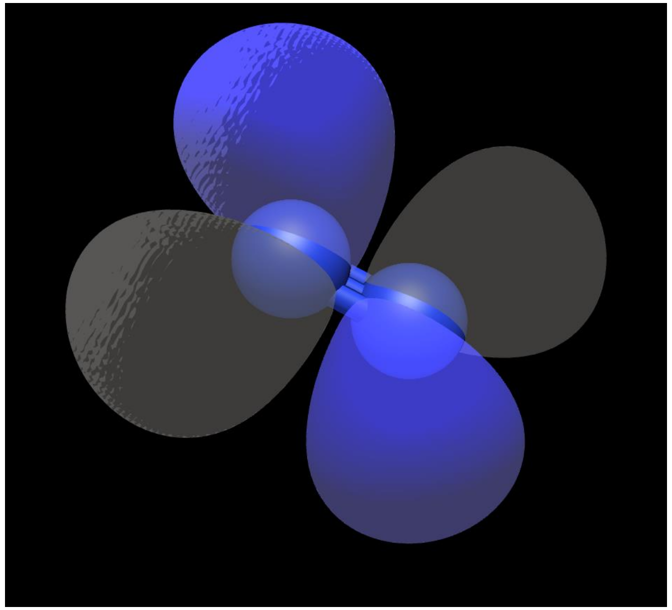


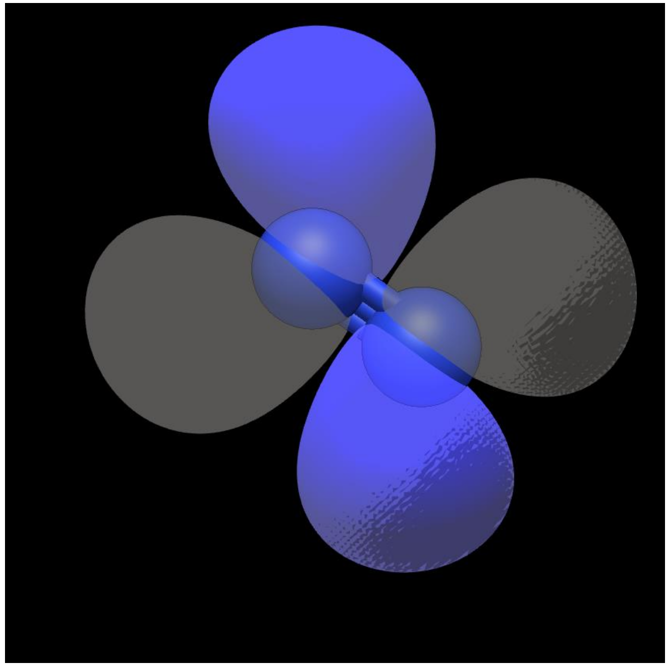


HOMO

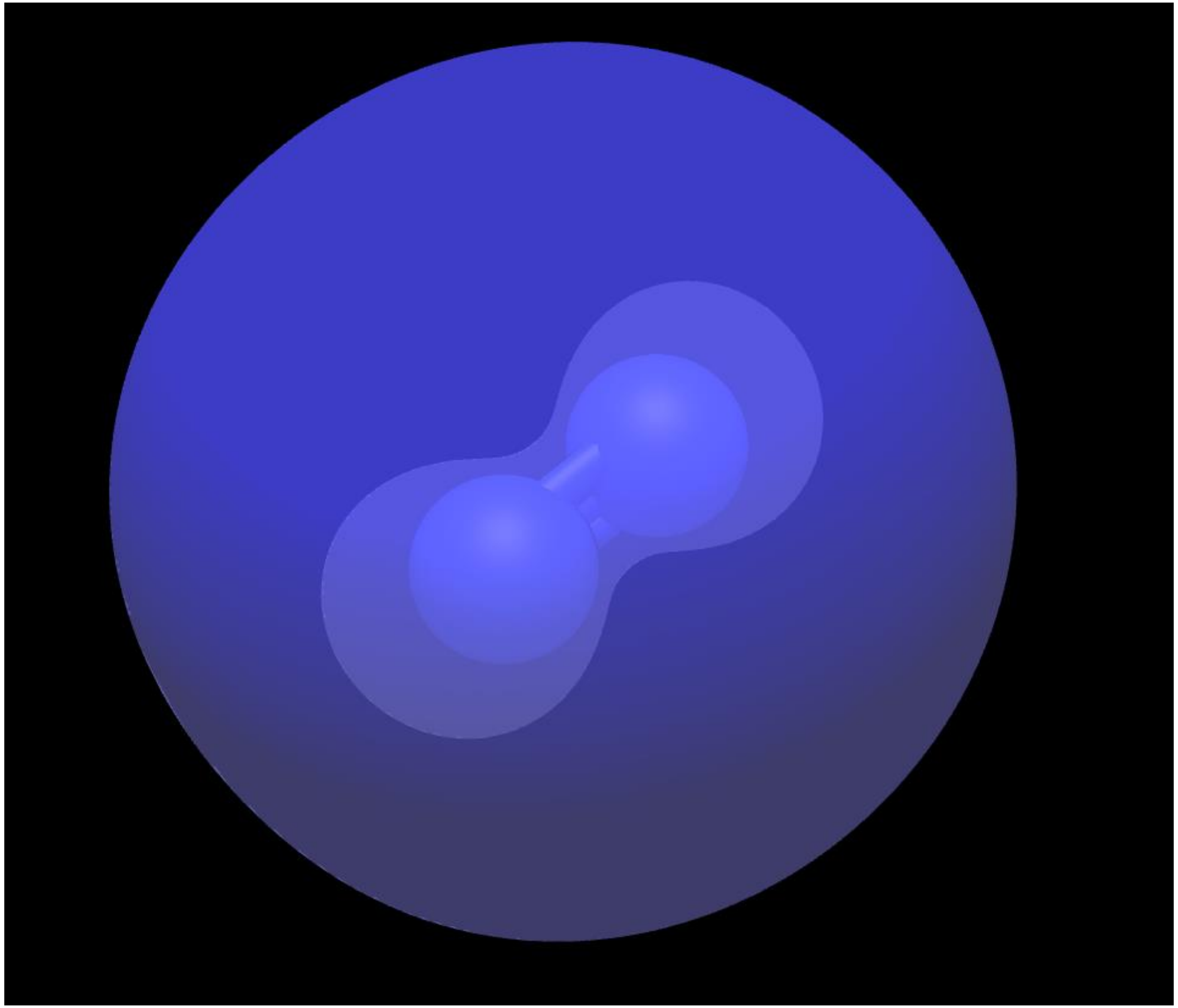


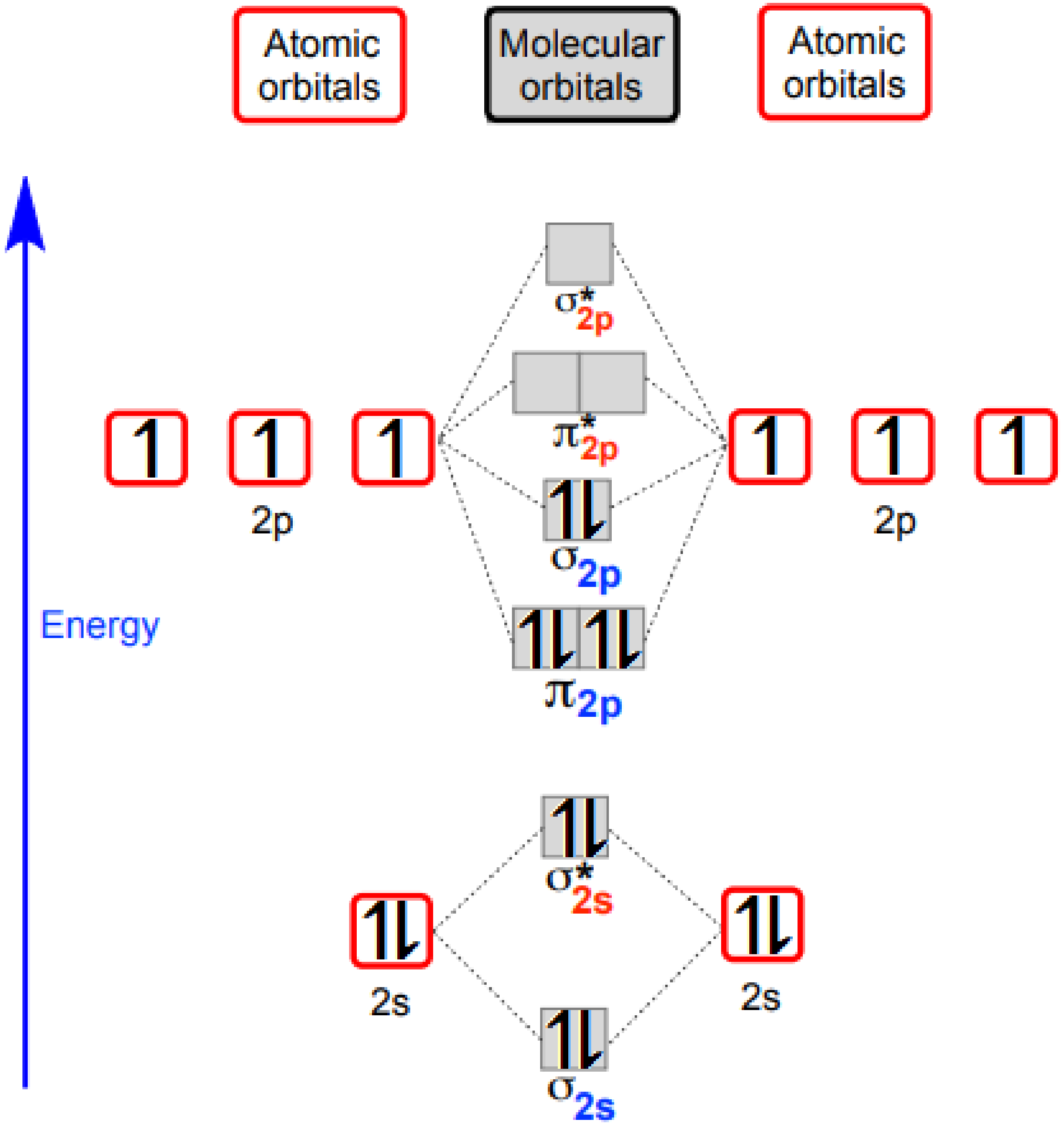
LUMO





Sigma*




Atomic orbitals
Molecular orbitals
Atomic orbitals
Energy
σ*2p
π*2p
σ2p
π2p
2p
2p
σ*2s
2s
2s
σ2s

Molecular Orbitals of O_2 using ORCA and their energies